

3rd Sem.-B.Tech.

END-SEMESTER EXAMINATION, JULY 2026 (SUMMER SEMESTER)

Course Code- EAMTC04/ EPMTTC301

Course Title- Mathematics for Machine Learning

Time- 03 Hours

Max. Marks-50

Note: - Attempt all five questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer.

Q.No.	Attempt any 2 parts of the questions below	Marks	CO
1(a)	Find Kernel and Image of the linear transformation $T: R^4 \rightarrow R^3$ such that $T(x, y, z, w) = (x - z, y + w, x + y + z)$. Also, verify the Rank-Nullity Theorem.	05	CO1
1(b)	A mapping $T: R^3 \rightarrow R^3$ defined by $T(x, y, z) = (2x - y + z, x + 3y + 2z, 3x + 2y + z)$. Show that T is a linear mapping. Find $\text{Ker}T$ and the dimension of $\text{Ker}T$.	05	CO1
1(c)	Let V and W be two vector spaces in R^3 . Let $T: V \rightarrow W$ be a linear transformation defined by $T(x, y, z) = (x + y, y + z, x + z)$. Find the matrix representation of T with respect to the basis $X = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ in V and $Y = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$ in W .	05	CO1
2(a)	Use the Gram-Schmidt process to convert the basis $\{(2, 1, 0), (1, 2, 1), (0, 1, 3)\}$ of the Euclidean space R^3 with standard inner product into an orthonormal basis.	05	CO2
2(b)	Find the eigenvalues and eigen vectors of $A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{bmatrix}$	05	CO2
2(c)	Reduce the quadratic form $Q = 3x^2 + 4y^2 + 5z^2 + 2xy + 2yz + 2zx$ to the normal form and check whether it is positive definite or positive semi-definite.	05	CO2
3(a)	Find the polar decomposition of $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.	05	CO3
3(b)	Find the pseudo-inverse of the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$.	05	CO3

3(c)	Find the singular value decomposition of the matrix $A = \begin{pmatrix} 2 & 0 & 2 \\ 2 & 0 & 2 \end{pmatrix}$.	05	CO3
4(a)	Verify Green's Lemma in the xy plane for $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the closed curve of the region bounded by $x = 0, y = 0, x + y = 1$.	05	CO4
4(b)	Show that the vector field $\vec{V} = 3x^2y^2z^4\hat{i} + 2x^3yz^4\hat{j} + 4x^3y^2z^3\hat{k}$ is irrotational. Hence find a scalar function $\phi(x, y, z)$ such that $V = \text{grad}\phi$	05	CO4
4(c)	If $\vec{F} = x\hat{i} + (z^2 - zx)\hat{j} - xy\hat{k}$. Evaluate $\iint_S \vec{F} \cdot \vec{ds}$ where S is the surface of the triangle with vertices (2,0,0), (0,2,0), (0,0,4).	05	CO4
5(a)	Examine if the set $S = \{(u, v) \in R^2: u^2 + v^2 \leq 25\}$ is a convex set.	05	CO5
5(b)	Maximize: $x^m y^n z^p$ Subject to: $x + y + z - a = 0$	05	CO5
5(c)	Solve: $\text{Min } f(x) = x_1^2 + x_2^2 - 6x_1 - 4x_2 + 13$ Subject to $x_1^2 + x_2^2 \leq 52$ $x_1, x_2 \geq 0$	05	CO5